PROBABILITY AND STATISTICS I

Sample Spaces

Lecture 02

Random experiments · events · the classical probability

Lecture notes with every worked example,
tree and grid diagrams, and practice problems for independent work.

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1 Random experiments

Definition 2.1 — Random experiment. An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a **random experiment**.

Typical examples of a random experiment are the roll of a die, the toss of a coin, and drawing a card from a deck.

The word doing the work in that definition is **even though**. Repeating the experiment identically is not enough to force the same outcome; that failure to repeat is not sloppiness in the setup, it is the phenomenon being studied. Everything in this chapter is machinery for describing it.

Note. Chapter 1 answered "how many?". This chapter asks "how many **out of** how many?" — and the counting techniques become the numerator and denominator of every probability you write down. Nothing from Chapter 1 is retired here; it is put to work.

2 Sample space

Definition 2.2 — Sample space. The set of all possible outcomes of a random experiment is called the **sample space** of the experiment. The sample space is denoted S or Ω.

Example 1. A coin is tossed once. Find the sample space.
Solution. The experiment has one stage with two possible outcomes, heads $\{H\}$ or tails $\{T\}$, and nothing else can happen. Listing them,

$$S = \{H, T\}, \quad N = |S| = 2.$$

Note. The same question is asked in several shapes, and they all want this answer: "What is the sample space of tossing a coin once?", "What is the sample space of tossing a regular coin?", "Let a coin be tossed once — find the sample space."

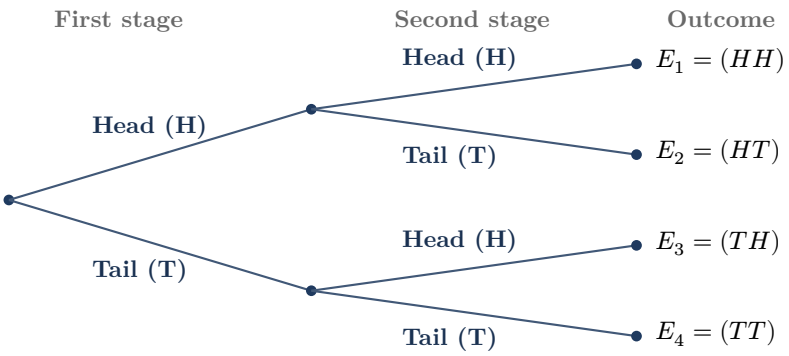
Example 2. Find the sample space for the experiment of tossing a coin twice.
(Equivalently: "find the sample space of the random experiment of flipping a coin twice", or "what is the sample space of tossing a regular coin two consecutive times?")
Solution. There are four outcomes. They are pairs of heads and tails, so

$$S = \{(H, H), (H, T), (T, H), (T, T)\},$$

which is more commonly written

$$S = \{HH, HT, TH, TT\}.$$

The four outcomes are exactly the four paths through a two-stage tree:



Watch out. (H, T) and (T, H) are **different outcomes** — they are different paths through the tree. Students who write $S = \{HH, HT, TT\}$ with three elements have quietly decided order does not matter, and every probability they compute afterwards will be wrong. The tree is the cure: count the paths, not the pictures.

Example 3. A die is tossed once. Find the sample space.

Solution. A die has six faces, and exactly one of them lands uppermost. The outcome is the number showing, so

$$S = \{1, 2, 3, 4, 5, 6\}, \quad N = 6.$$

("die" and "dice" are used interchangeably in these questions; "dice" is strictly the plural.)

Example 4. Find the sample space for the experiment of tossing a die twice.

Solution. The first die may show any of six numbers and the second die may show any one of six numbers. By the multiplication rule (Rule 1.1 of Chapter 1) there are $6 \times 6 = 36$ outcomes:

1st \ 2nd	**1**	**2**	**3**	**4**	**5**	**6**
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

The outcomes may equally be represented by a tree with 36 branches — but at this size the grid is the better picture, and it is the one to keep: every two-dice question for the rest of the course is a matter of shading cells in it.

Example 5. What is the sample space when a ball is selected from a box containing 3 symmetric balls (red, yellow, green)?

Solution. One ball is drawn, and the three balls are distinguishable only by colour — so the outcome of the experiment **is** the colour drawn. There are three of them:

$$S = \{\text{Red}, \text{Yellow}, \text{Green}\}, \quad N = 3.$$

"Symmetric" is doing a job here: it is the word that makes the three outcomes equally likely, which is what Rule 2.1 in §4 will need.

Example 6. What is the sample space of the 2-digit numbers formed using the digits $\{2, 3\}$?

Solution. Two slots to fill — tens and units — with 2 choices for each, and nothing forbids using the same digit twice. By the multiplication rule there are $2 \times 2 = 4$ outcomes:

$$\boxed{2} \times \boxed{2}$$

Listing them,

$$S = \{22, 33, 23, 32\}, \quad N = 4.$$

Both 23 and 32 appear, because the slots are different positions: order matters here.

Example 7. An experiment consists of flipping a coin and then flipping it a second time if a head occurs; if a tail occurs on the first flip, then a die is tossed once. Find the sample space.
Solution. Split on the first flip, because the experiment itself does:

- **If the first flip is a head** (H), the coin is flipped again — 2 outcomes, HH and HT.
- **If the first flip is a tail** (T), a die is tossed — 6 outcomes, $T1$ through $T6$.

The two branches cannot both happen, so the sample space is their union and the counts add:

$$S = \{HH, HT\} \cup \{T1, T2, T3, T4, T5, T6\} = \{HH, HT, T1, T2, T3, T4, T5, T6\},$$

with $N = 2 + 6 = 8$.

Note. Example 7 is the one worth slowing down on. The second stage **is not the same** on both branches — two outcomes after a head, six after a tail — so the multiplication rule does not apply and $2 \times 2 = 4$ is not the answer. You add the branches instead: $2 + 6 = 8$. Draw the tree and the structure is obvious; write formulas first and it is invisible.

Definition 2.3 — Discrete and continuous sample spaces. A sample space is **discrete** if it consists of a finite or countably infinite set of outcomes. A sample space is **continuous** if it contains an interval — either finite or infinite — of real numbers.

Every sample space in this chapter is discrete and finite, which is what makes counting the right tool. Continuous sample spaces arrive later, with the normal distribution, and they need integrals rather than counts.

Exercise set 1

1.1 Write the sample space for tossing a coin three times. How many outcomes?

1.2 A coin is tossed and a die is rolled. Write S, and say why it has 12 elements.

1.3 An experiment rolls a die; if the result is 6 the die is rolled again, otherwise a coin is tossed. Write S and count it.

1.4 Is the sample space of "the time in seconds until the next bus arrives" discrete or continuous? Justify in one sentence.

3 Events

Definition 2.4 — Event. An **event** is a subset of the sample space of a random experiment.

Example 8. Consider the experiment of Example 3 — a die tossed once. These are some events:

A: observe an even number
B: observe a multiple of 3
C: observe a 5
D: observe a 7

Solution. Since $S = \{1, 2, 3, 4, 5, 6\}$,

$$A = \{2, 4, 6\}, \quad B = \{3, 6\}, \quad C = \{5\}, \quad D = \Phi,$$

where Φ is the empty set — the set with no elements in it.

Note. $D = \Phi$ is not a failure of the question. The empty set is a perfectly good event; it is the event that cannot happen, and in §4 it will earn probability 0. Likewise S itself is an event — the one that always happens.

Example 9. If a die is tossed once, then $S = \{1, 2, 3, 4, 5, 6\}$, and

$$A_1 = \{1, 3, 5\} \text{ is an event}, \qquad A_2 = \{2, 5, 7\} \text{ is not an event.}$$

Solution. $A_1 \subseteq S$, so it is an event. A_2 contains 7, which is not a possible outcome of this experiment, so $A_2 \not\subseteq S$ and A_2 is not an event of it.

Example 10. List the elements of each of the following sets.

(i) the set of integers **between** 1 and 30;
(ii) the set of integers **from** 1 to 30;
(iii) the set of integers between 1 and 30 which are divisible by 4;
(iv) the set $S = \{x : x^2 + 2x - 3 = 0\}$;
(v) the set $S = \{x : 3x - 9 \geq 0 \text{ and } x < 2\}$.

Solution. (i) $S = \{2, 3, 4, 5, 6, ..., 29\}$ — "between" excludes both endpoints.

(ii) $S = \{1, 2, 3, 4, 5, ..., 30\}$ — "from ... to" includes them.

(iii) $S = \{4, 8, 12, 16, 20, 24, 28\}$.

(iv) Factorise:

$$x^2 + 2x - 3 = 0 \Rightarrow (x - 1)(x + 3) = 0 \Rightarrow S = \{1, -3\}.$$

(v) Solve the two conditions and intersect them:

$$3x - 9 \geq 0 \text{ and } x < 2 \Rightarrow 3x \geq 9 \text{ and } x < 2 \Rightarrow x \geq 3 \text{ and } x < 2.$$

No number satisfies both, so $S = \{\} = \Phi$.

Watch out. Parts (i) and (ii) differ by one word and by two elements. "Between" is exclusive, "from ... to" is inclusive — and since these sets become denominators in §4, the word changes the probability. Read the preposition before you count.

Exercise set 2

2.1 For a die tossed once, write the events "an odd number", "a number at least 5", and "a number less than 1" as subsets of S.

2.2 For a coin tossed twice, write the event "at least one head" as a subset, and the event "exactly one head".

2.3 A sample space has n outcomes. How many different events does it have? (Hint: each outcome is either in a given subset or not.)

2.4 List the elements of $\{x : x^2 - 5x + 6 = 0\}$ and of $\{x : x \text{ is an integer}, 2 < x \leq 6\}$.

4 The concept of probability

If we are sure or certain that an event will occur, we say its probability is 100%, or 1. If we are sure the event will not occur, we say its probability is zero. If, for example, the probability is $\frac{1}{4}$, we would say there is a 25% chance it will occur and a 75% chance it will not. Equivalently, the odds against occurrence are 75% to 25%, or 3 to 1.

Rule 2.1 — Classical (equally likely) definition of probability.

If an experiment can result in any one of N different **equally likely** outcomes, and if exactly n of these outcomes correspond to event A, then the probability of event A is

$$P(A) = \frac{n}{N},$$

where $P(A)$ is called the probability of the event A.

Watch out. The two words that carry the whole rule are **equally likely**. $P(A) = n/N$ is not a definition of probability in general — it is what probability reduces to when the outcomes are symmetric. A loaded die still has $N = 6$ and $n = 3$ for "even", and $P(\text{even})$ is still not $1/2$. Say out loud why the outcomes are equally likely before you divide.

Three consequences fall straight out of the rule, and they are worth stating once:

Fact	Why
$0 \leq P(A) \leq 1$ for every event A	n is a count of a subset of N outcomes, so $0 \leq n \leq N$.
$P(\Phi) = 0$	The empty event contains no outcomes: $n = 0$.
$P(S) = 1$	S contains all of them: $n = N$.

Note. Use the first line as a reflex check on every answer for the rest of the term. A probability above 1 or below 0 is not a small slip to be corrected quietly — it means the model was wrong, and it is free to catch.

Example 11. From Example 8, find $P(A)$, $P(B)$, $P(C)$ and $P(D)$.
Solution. All six faces are equally likely, so $N = 6$ and only the counts change:

$$P(A) = \frac{3}{6} = \frac{1}{2}, \quad P(B) = \frac{2}{6} = \frac{1}{3}, \quad P(C) = \frac{1}{6}, \quad P(D) = \frac{0}{6} = 0.$$

Example 12. A container has 5 red balls, 10 white balls and 35 blue balls. One of the balls is selected randomly. Find the probability that the selected ball is (A) red (B) white (C) blue.

Solution. This experiment has $5 + 10 + 35 = 50$ equally likely outcomes. Using the classical definition,

$$P(A) = \frac{5}{50} = 0.1, \quad P(B) = \frac{10}{50} = 0.2, \quad P(C) = \frac{35}{50} = 0.7.$$

As a check, $0.1 + 0.2 + 0.7 = 1$: the three colours exhaust the box, so their probabilities must sum to 1.

Example 13. The experiment of selecting one card randomly from a standard deck has 52 equally likely outcomes. Find the probabilities of

$$A = \{\text{club}\}, \quad B = \{\text{red card}\}, \quad C = \{\text{face card (Jack, Queen, King)}\}.$$

Solution. A deck has four suits of 13 cards, two suits red and two black, and three face cards per suit:

$$P(A) = \frac{13}{52} = 0.25, \quad P(B) = \frac{26}{52} = 0.5, \quad P(C) = \frac{12}{52} \approx 0.23.$$

4.1 The method, in three lines

Every question in this chapter, and every question in Assignment 2, is the same three steps.

Step	What you write	Where it goes wrong
1. Describe the experiment and its sample space	$S = \ldots$, and $N = \lvert S \rvert$	Counting S under a different reading of the question than the one intended
2. Write the event as a subset	$A = \ldots$, and $n = \lvert A \rvert$	Vague wording left in English — "at least one" is not a subset until you list it
3. Divide, then check	$P(A) = n/N$, and $0 \le P(A) \le 1$	Outcomes that were never equally likely in the first place

Note. When "at least one" appears, reach for the complement: $P(A) = 1 - P(A^c)$. "At least one head in three tosses" has seven outcomes to list and one to subtract. Chapter 1's exercise 3.5 made the same point about committees; it is the same move.

Exercise set 3

3.1 A die is rolled. Find P(the number is greater than 4) and P(the number is prime).

3.2 Two coins are tossed. Find P(exactly one head) and P(at least one head).

3.3 Two dice are rolled. Use the grid of Example 4 to find $P(\text{sum} = 7)$ and $P(\text{sum} \le 4)$.

3.4 A jar has 3 red, 7 green and 10 white marbles. One is drawn. Find P(white) and P(not white), and check they sum to 1.

3.5 A card is drawn from a standard deck. Find P(a queen) and P(the 3 of diamonds).

3.6 Three coins are tossed. Find P(all heads) and P(at least one head) — the second by complement.